

Module 2: Lab00

Implementing a Basic Web Server

Grading Criteria

Total Points: 70 + 30 extra credit points available

You may only use your own original Python libraries and the thread and socket Python libraries.

☐ [2.5 points] Your submission is labeled as “cs460_module02_lab00_[nau_id]_[lastname]_[firstname].zip.” For example, if I were to submit a file, it would be labeled as cs460_module02_lab00_mv668_vigil-hayes_morgan.zip. **FAILURE TO COMPLY WITH THIS STEP CAN LEAD TO A ZERO GRADE.**

☐ [7.5 points] Your submission files are in the correct format. **FAILURE TO COMPLY WILL RESULT IN A ZERO GRADE.**

- ☐ [2.5 points] All files must be submitted as a single ZIP file
- ☐ [2.5 points] Server code is all contained in a file called [nau_id]_web_server.py
- ☐ [2.5 points] This lab manual is contained in a file called [nau_id]_lab_manual.pdf

☐ [20 points] Code runs without errors in Unix environment and is highly readable. I will be testing code on Mac OS X and Ubuntu environments.

- ☐ [5 points] XX_web_server.py starts without error and is able to start up a main server thread that handles incoming client requests.
- ☐ [5 points] XX_web_server.py runs using the following command at the terminal:

```
python3 XX_web_server.py [socket]
```
- ☐ [5 points] Code is well organized into subroutines (helper functions) that
- ☐ [5 points] I am able to use a simple telnet client to test the server

☐ [20 points] Your server correctly complies with RFC 2616

- ☐ [5 points] When the server receives a correctly formatted GET request for the file at the root directory, the server is able to provide a correctly formatted response that results in sending a small bit of HTML with an original message embedded (e.g., “Computer Networks are the best!”)
- ☐ [5 points] When the server receives correctly formatted HTTP requests that are NOT GET requests, the server follows RFC 2616 and sends the proper response indicating a lack of server implementation support.

- ☐ [5 points] When the server receives correctly formatted HTTP requests that is a GET request for a file that does not exist in the server's directory, the server needs to send the proper response according to RFC 2616.
- ☐ [5 points] When the server receives an incorrectly formatted HTTP request, it needs to respond according to RFC 2616.

☐ [20 points] Part 1 of the Lab Manual (i.e., this document) has been completed correctly and completely

- ☐ [2.5 points] Identify where in RFC 2616 you found the rules for correctly responding to a correctly formatted GET request for the root directory
- ☐ [2.5 points] Show a screenshot of a Wireshark capture of the GET request packet sent to your server and the server response
- ☐ [2.5 points] Identify where in RFC 2616 you found the rules for correctly responding to a correctly formatted GET request for a nonexistent file
- ☐ [2.5 points] Show a screenshot of a Wireshark capture of the GET request for nonexistent file packet sent to your server and the server response
- ☐ [2.5 points] Identify where in RFC 2616 you found the rules for correctly responding to correctly formatted non-GET requests (e.g., POST) for a nonexistent file
- ☐ [2.5 points] Show a screenshot of a Wireshark capture of the correctly formatted non-GET request packet sent to your server and the server response
- ☐ [2.5 points] Identify where in RFC 2616 you found the rules for correctly responding to incorrectly formatted requests
- ☐ [2.5 points] Show a screenshot of a Wireshark capture of the incorrectly formatted request packet sent to your server and the server response

Part 1: Simple Web Server

[2.5 points] Identify where in RFC 2616 you found the rules for correctly responding to a correctly formatted GET request for the root directory

10.2.1 200 OK

The request has succeeded. The information returned with the response is dependent on the method used in the request, for example:

GET an entity corresponding to the requested resource is sent in the response;

HEAD the entity-header fields corresponding to the requested resource are sent in the response without any message-body;

POST an entity describing or containing the result of the action;

[2.5 points] Show a screenshot of a Wireshark capture of the GET request packet sent to your server and the server response

capture00.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 2080

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	127.0.0.1	127.0.0.1	TCP	80	43564 → 2080 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=2406682998 TSecr=0 WS=128
2	0.000011	127.0.0.1	127.0.0.1	TCP	80	2080 → 43564 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=2406682998 TSecr=2406682998
3	0.000021	127.0.0.1	127.0.0.1	TCP	72	43564 → 2080 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=2406682998 TSecr=2406682998
4	0.000082	127.0.0.1	127.0.0.1	HTTP	107	GET / HTTP/1.1
5	0.000087	127.0.0.1	127.0.0.1	TCP	72	2080 → 43564 [ACK] Seq=1 Ack=36 Win=65536 Len=0 TSval=2406682998 TSecr=2406682998
6	0.000250	127.0.0.1	127.0.0.1	HTTP	345	HTTP/1.1 200 OK (text/html)
7	0.000271	127.0.0.1	127.0.0.1	TCP	72	43564 → 2080 [ACK] Seq=36 Ack=274 Win=65280 Len=0 TSval=2406682998 TSecr=2406682998
8	0.000287	127.0.0.1	127.0.0.1	TCP	72	2080 → 43564 [FIN, ACK] Seq=274 Ack=36 Win=65536 Len=0 TSval=2406682998 TSecr=2406682998
9	0.000323	127.0.0.1	127.0.0.1	TCP	72	43564 → 2080 [FIN, ACK] Seq=36 Ack=275 Win=65536 Len=0 TSval=2406682998 TSecr=2406682998
10	0.000336	127.0.0.1	127.0.0.1	TCP	72	2080 → 43564 [ACK] Seq=275 Ack=37 Win=65536 Len=0 TSval=2406682998 TSecr=2406682998

[2.5 points] Identify where in RFC 2616 you found the rules for correctly responding to a correctly formatted GET request for a nonexistent file

10.4.5 404 Not Found

The server has not found anything matching the Request-URI. No indication is given of whether the condition is temporary or permanent. The 410 (Gone) status code SHOULD be used if the server knows, through some internally configurable mechanism, that an old resource is permanently unavailable and has no forwarding address. This status code is commonly used when the server does not wish to reveal exactly why the request has been refused, or when no other response is applicable.

[2.5 points] Show a screenshot of a Wireshark capture of the GET request for nonexistent file packet sent to your server and the server response

1	0.000000	127.0.0.1	127.0.0.1	TCP	80	42842 → 2080 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=2408087059 TSecr=0 WS=128
2	0.000012	127.0.0.1	127.0.0.1	TCP	80	2080 → 42842 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=2408087059 TSecr=2408087059
3	0.000021	127.0.0.1	127.0.0.1	TCP	72	42842 → 2080 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=2408087059 TSecr=2408087059
4	0.000085	127.0.0.1	127.0.0.1	HTTP	116	GET /nope.html HTTP/1.1
5	0.000091	127.0.0.1	127.0.0.1	TCP	72	2080 → 42842 [ACK] Seq=1 Ack=45 Win=65536 Len=0 TSval=2408087059 TSecr=2408087059
6	0.000186	127.0.0.1	127.0.0.1	HTTP	294	HTTP/1.1 404 Not Found (text/html)
7	0.000207	127.0.0.1	127.0.0.1	TCP	72	42842 → 2080 [ACK] Seq=45 Ack=223 Win=65408 Len=0 TSval=2408087059 TSecr=2408087059
8	0.000221	127.0.0.1	127.0.0.1	TCP	72	2080 → 42842 [FIN, ACK] Seq=223 Ack=45 Win=65536 Len=0 TSval=2408087059 TSecr=2408087059
9	0.000253	127.0.0.1	127.0.0.1	TCP	72	42842 → 2080 [FIN, ACK] Seq=45 Ack=224 Win=65536 Len=0 TSval=2408087059 TSecr=2408087059
10	0.000264	127.0.0.1	127.0.0.1	TCP	72	2080 → 42842 [ACK] Seq=224 Ack=46 Win=65536 Len=0 TSval=2408087059 TSecr=2408087059

[2.5 points] Identify where in RFC 2616 you found the rules for correctly responding to correctly formatted non-GET requests (e.g., POST) for a nonexistent file

10.5.2 501 Not Implemented

The server does not support the functionality required to fulfill the request. This is the appropriate response when the server does not recognize the request method and is not capable of supporting it for any resource.

[2.5 points] Show a screenshot of a Wireshark capture of the correctly formatted non-GET request packet sent to your server and the server response

1	0.000000	127.0.0.1	127.0.0.1	TCP	80	35754 → 2080 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=2408119044 TSecr=0 WS=128
2	0.000011	127.0.0.1	127.0.0.1	TCP	80	2080 → 35754 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=2408119044 TSecr=2408119044
3	0.000020	127.0.0.1	127.0.0.1	TCP	72	35754 → 2080 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=2408119044 TSecr=2408119044
4	0.000079	127.0.0.1	127.0.0.1	HTTP	108	POST / HTTP/1.1
5	0.000084	127.0.0.1	127.0.0.1	TCP	72	2080 → 35754 [ACK] Seq=1 Ack=37 Win=65536 Len=0 TSval=2408119044 TSecr=2408119044
6	0.000171	127.0.0.1	127.0.0.1	HTTP	306	HTTP/1.1 501 Not Implemented (text/html)
7	0.000192	127.0.0.1	127.0.0.1	TCP	72	35754 → 2080 [ACK] Seq=37 Ack=235 Win=65408 Len=0 TSval=2408119044 TSecr=2408119044
8	0.000206	127.0.0.1	127.0.0.1	TCP	72	2080 → 35754 [FIN, ACK] Seq=235 Ack=37 Win=65536 Len=0 TSval=2408119044 TSecr=2408119044
9	0.000235	127.0.0.1	127.0.0.1	TCP	72	35754 → 2080 [FIN, ACK] Seq=37 Ack=236 Win=65536 Len=0 TSval=2408119044 TSecr=2408119044
10	0.000246	127.0.0.1	127.0.0.1	TCP	72	2080 → 35754 [ACK] Seq=236 Ack=38 Win=65536 Len=0 TSval=2408119044 TSecr=2408119044

[2.5 points] Identify where in RFC 2616 you found the rules for correctly responding to incorrectly formatted requests

10.4.1 400 Bad Request

The request could not be understood by the server due to malformed syntax. The client SHOULD NOT repeat the request without modifications.

[2.5 points] Show a screenshot of a Wireshark capture of the incorrectly formatted request packet sent to your server and the server response

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	127.0.0.1	127.0.0.1	TCP	80	51156 → 2080 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=2408340714 TSecr=0 WS=128
2	0.000014	127.0.0.1	127.0.0.1	TCP	80	2080 → 51156 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=2408340714 TSecr=24083...
3	0.000027	127.0.0.1	127.0.0.1	TCP	72	51156 → 2080 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=2408340714 TSecr=2408340714
4	0.000101	127.0.0.1	127.0.0.1	TCP	86	51156 → 2080 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=14 TSval=2408340714 TSecr=2408340714
5	0.000108	127.0.0.1	127.0.0.1	TCP	72	2080 → 51156 [ACK] Seq=1 Ack=15 Win=65536 Len=0 TSval=2408340714 TSecr=2408340714
6	0.000177	127.0.0.1	127.0.0.1	HTTP	298	HTTP/1.1 400 Bad Request (text/html)
7	0.000199	127.0.0.1	127.0.0.1	TCP	72	51156 → 2080 [ACK] Seq=15 Ack=227 Win=65408 Len=0 TSval=2408340714 TSecr=2408340714
8	0.000213	127.0.0.1	127.0.0.1	TCP	72	2080 → 51156 [FIN, ACK] Seq=227 Ack=15 Win=65536 Len=0 TSval=2408340714 TSecr=2408340714
9	0.000239	127.0.0.1	127.0.0.1	TCP	72	51156 → 2080 [FIN, ACK] Seq=15 Ack=228 Win=65536 Len=0 TSval=2408340714 TSecr=2408340714
10	0.000249	127.0.0.1	127.0.0.1	TCP	72	2080 → 51156 [ACK] Seq=228 Ack=16 Win=65536 Len=0 TSval=2408340714 TSecr=2408340714